



Project Report

➤ **Theoretical Explanation of The Project**

This project creates an autonomous, smart environment for a lecture hall. The core philosophy is that the system must operate independently of any external server; it performs self-diagnostics at startup and manages the hall's environment (Lighting and Climate) based on real-time occupancy and environmental data. While Node-RED provides a "Digital Twin" for remote monitoring and manual override via a secure login, the ESP32 remains the primary decision-maker to ensure the hall remains functional even if network communication is lost.

➤ **All Components Organized**

The system hardware is composed of the following units

- **Controller:** ESP32 Microcontroller.
- **Input Sensors:** DHT22 (Temperature/Humidity), MQ-2 (Gas/Smoke), HC-SR501 (PIR Motion), and HC-SR04 (Ultrasonic).
- **Visual Outputs:** I2C LCD Screen (16x2), 1 Red LED, 1 Yellow LED, 1 Green LED, 1 Blue LED (PWM), and 4 White LEDs.
- **Auditory Output:** Active Buzzer.
- **Support Hardware:** Resistors for all LEDs and I2C pull-ups..

➤ **The Role of Each Component**

- **ESP32:** Manages all logic, processes sensor data locally, and syncs with Node-RED.
- **DHT22:** Measures temperature and humidity for climate control logic.
- **MQ-2:** Monitors air safety to detect fire or gas leaks.
- **HC-SR04:** Operates as a entry counter for the first 2 minutes of the system cycle.
- **HC-SR501:** Monitors continuous motion to confirm if the hall is currently in use.
- **I2C LCD:** Displays system status, sensor readings, and error messages locally.
- **LEDs:** Represent system states (Red/Yellow/Green) and simulated appliances (Blue for AC, 1 White for Fan, 3 White for Lighting).
- **Buzzer:** Provides audible alerts for hardware actions and emergency alarms.

➤ **The Master Wiring Checklist (Direct Pin-to-Pin)**

- **Power Rails (Breadboard)**
 - GND Rail (Blue/Black): Connected to ESP32 GND.
 - 3.3V Rail (Red): Connected to ESP32 3V3.
 - 5V Rail (Red): Connected to ESP32 VIN.

- **Input Sensors (Direct Data Connections)**

Component	Function	ESP32 Pin (Label)	Power Source
DHT22	Temp/Humidity	D4	5V (VIN)
MQ-2	Gas/Smoke	D34	
HC-SR501	Motion (PIR)	D27	
Ultrasonic	Trig	D12	
Ultrasonic	Echo	D14	

- **Output Components (Direct Connections)**

Component	Function	ESP32 Pin (Label)	Power Source
I2C LCD	SDA	D21	5V (VIN)
I2C LCD	SCL	D22	5V (VIN)
Active Buzzer	Alarm	D13	Direct Pin
Green LED	System OK	D2	With Resistor
Yellow LED	Warning/Fault	D5	With Resistor
Red LED	Fire/Emergency	D15	With Resistor
Blue LED	AC (Cooling)	D25	With Resistor
White LED 1	Ventilation Fan	D26	With Resistor
White LED 2	Light Zone 1	D23	With Resistor
White LED 3	Light Zone 2	D19	With Resistor
White LED 4	Light Zone 3	D18	With Resistor

➤ **All Possible System Scenarios**

System Initialization and Self-Check

When the ESP32 is powered on, the system begins by scanning all connected sensors and displays "Scanning System" on the LCD screen. During the scanning process, the name of each sensor is shown along with its status (Working / Not working).

If any sensor is disconnected or malfunctioning, a warning message "Check Sensor" is displayed with the sensor name at the end of the scan, confirming the issue after completing the full system check. In this case, the buzzer triggers a single alert sound, and the yellow LED turns on and continues blinking repeatedly until the problem is resolved.

If all components are working correctly, The Green LED turns on to indicate that the system is in normal operating mode. The buzzer produces two beeps at startup, and after the initialization is completed, it plays a soft and pleasant tone to indicate successful completion.

System Status Indicators

The system uses LEDs for status indication. The green LED indicates that the system is running. The yellow LED blinks if there is a problem (Such as a Disconnected Sensor), while the system continues operating. The red LED is reserved for critical alerts such as fire or gas leakage.

Dual Mode Operation (Online & Offline)

The system operates in two modes: online and offline. In online mode, the ESP32 sends sensor data to Node-RED using MQTT, allowing real-time monitoring and full user control. In offline mode, the system works autonomously using predefined automation logic without user intervention, ensuring continuous operation even without internet connectivity.

Internet Connectivity Handling

After initialization, the system attempts to connect to a Wi-Fi network. If successful, it switches to online mode and starts sending data to Node-RED. If the connection fails, the system continues in offline automation mode. If the connection drops during operation, the system automatically switches to offline mode and reconnects once the network is available again without interrupting functionality.

Idle Mode (No Occupancy)

When no students are detected, the system enters idle mode. In this state, only essential sensors (PIR, Ultrasonic, Temperature, Humidity, and Smoke sensors) remain active. The LCD displays temperature and humidity readings only. No actuators (Lights, Fan, or AC) are activated to conserve energy.

Intelligent Occupancy Detection

The ultrasonic sensor measures distance to detect entry and exit at the door, while the PIR sensor confirms motion inside the room. When a student enters, the system detects a decrease in distance and confirms motion, then increments the occupancy count. The LCD displays "Motion Detected," and the system becomes active.

Adaptive Lighting System

Lighting is controlled based on occupancy only. As the number of students increases, more white LEDs are activated to increase brightness (1 LED For Low Occupancy, Up To 3 LEDs For Maximum). Lighting decreases automatically as the number of students decreases.

Temperature-Based Climate Control

The system controls cooling based on temperature readings. If the temperature is between 20°C and 27°C, a fan (Represented by a White LED) is activated. If the temperature exceeds 27°C, the air conditioning (Blue LED) is activated with increasing intensity. If the temperature drops below 20°C, All Cooling Systems are Turned off.

Humidity Control Mechanism

If humidity levels drop below 40% or rise above 70%, the system considers the environment unsuitable. In this case, a warning message is displayed on the LCD, and the fan is activated automatically to regulate the environment and restore normal conditions.

Ultrasonic Time-Based Operation

To optimize power consumption, the ultrasonic sensor operates only during specific time windows. It counts students during the first period (Entry Phase), then pauses, and later reactivates to detect exits. The system calculates occupancy by comparing entry and exit counts.

Dynamic Occupancy Tracking Logic

The system distinguishes between entry and exit based on timing and PIR confirmation. If motion is detected after a time gap, it is considered a new entry. Otherwise, it is treated as an exit. This ensures accurate occupancy tracking.

Buzzer Feedback System

The buzzer provides audible feedback for system actions. A single beep indicates an action (Such as Turning on a Light), double beeps indicate warnings, and continuous alerts indicate critical conditions like Fire.

Smoke and Gas Detection (Safety System)

If gas levels exceed a certain threshold (e.g., 1500), the system enters warning mode, where the red LED blinks and the buzzer produce a single alert sound. If gas levels reach a critical threshold (e.g., 3000), the system enters emergency mode, where all operations are stopped, and continuous alerts are activated while displaying "Fire Alarm" or "Gas Leak" on the screen.

Emergency Handling and Recovery

In online mode, the emergency state can be manually reset via Node-RED. In offline mode, the system automatically returns to normal operation once gas levels return to safe values. A message "System Work" is displayed upon recovery.

Periodic System Check (Fault Detection)

The system performs periodic checks (e.g., Every 4 Minutes) to ensure all components are functioning. If a fault is detected, the user is notified via the LCD and yellow LED.

Manual Control via Node-RED

In online mode, the user has full control over system outputs such as lighting, fan, and AC through Node-RED, regardless of occupancy. Sensor reading remains automatic, while outputs can be overridden by the user if needed.

➤ Screenshot Phase (Arduino - Node-RED)

Arduino IDE:

```
=====System Starts=====
Check Sensors
DHT22: Ok
MQ-2: Ok
PIR: Ok
Ultrasonic: Ok (Normal)
All Sensors Ok
ESP32 WiFi Test
Connecting to WiFi...
Connected to WiFi!
IP Address: 10.86.56.173
==== System Ready ====
Attempting MQTT connection...Connected to MQTT Broker!
```

```
Attempting MQTT connection...Connected to MQTT Broker!
System Update -> Temp: 25.70 C | Hum: 52.10 % | Gas: 2384
System Update -> Temp: 25.70 C | Hum: 52.30 % | Gas: 2198
System Update -> Temp: 25.80 C | Hum: 53.00 % | Gas: 2038
System Update -> Temp: 25.80 C | Hum: 53.20 % | Gas: 1835
System Update -> Temp: 25.80 C | Hum: 52.50 % | Gas: 1677
System Update -> Temp: 25.80 C | Hum: 52.70 % | Gas: 1554
System Update -> Temp: 25.80 C | Hum: 52.90 % | Gas: 1456
Ultrasonic: 1 Student. Entering ENTRY Phase
Light Zone One | Count: 1
Fan Started | Temp 20-27C
System Update -> Temp: 25.80 C | Hum: 52.10 % | Gas: 1358
System Update -> Temp: 25.80 C | Hum: 51.70 % | Gas: 1202
Student Entered. Count: 2
Light Zone One | Count: 2
```

Entering PAUSE Phase (Sensors OFF)

PAUSE Phase | Sensors OFF

System Update -> Temp: 25.80 C | Hum: 50.20 % | Gas: 775
System Update -> Temp: 25.90 C | Hum: 50.20 % | Gas: 752
System Update -> Temp: 25.90 C | Hum: 50.70 % | Gas: 749
System Update -> Temp: 25.80 C | Hum: 50.80 % | Gas: 751
System Update -> Temp: 25.80 C | Hum: 50.60 % | Gas: 727
System Update -> Temp: 25.90 C | Hum: 50.40 % | Gas: 720
System Update -> Temp: 25.80 C | Hum: 50.50 % | Gas: 683
System Update -> Temp: 25.80 C | Hum: 50.70 % | Gas: 711
System Update -> Temp: 25.90 C | Hum: 50.80 % | Gas: 686
System Update -> Temp: 25.80 C | Hum: 50.90 % | Gas: 688
System Update -> Temp: 25.90 C | Hum: 50.80 % | Gas: 687
System Update -> Temp: 25.80 C | Hum: 50.70 % | Gas: 663
System Update -> Temp: 25.80 C | Hum: 50.60 % | Gas: 656

EXIT Phase | Listening...

System Update -> Temp: 25.80 C | Hum: 50.80 % | Gas: 624
System Update -> Temp: 25.90 C | Hum: 50.80 % | Gas: 594
Student Exited. Total inside: 2
Light Zone One | Count: 2
Student Exited. Total inside: 1
Light Zone One | Count: 1
System Update -> Temp: 25.90 C | Hum: 50.90 % | Gas: 575
Student Exited. Total inside: 0
Room Empty. Returning to IDLE Mode
Room Empty | IDLE Mode
System Update -> Temp: 25.90 C | Hum: 51.10 % | Gas: 583
Cooling OFF | Room Empty
PIR: Motion Detected Once
System Update -> Temp: 25.80 C | Hum: 50.70 % | Gas: 581

Fan LOCKED ON - Gas Warning Active (Ventilation)

WARNING! | High Gas Level

Live Gas Level: 1808

Live Gas Level: 2015

Live Gas Level: 2219

Live Gas Level: 2431

Live Gas Level: 2634

Live Gas Level: 2836

WARNING! | High Gas Level

Live Gas Level: 3037

EMERGENCY: Gas Leak / Fire | Level: 3037

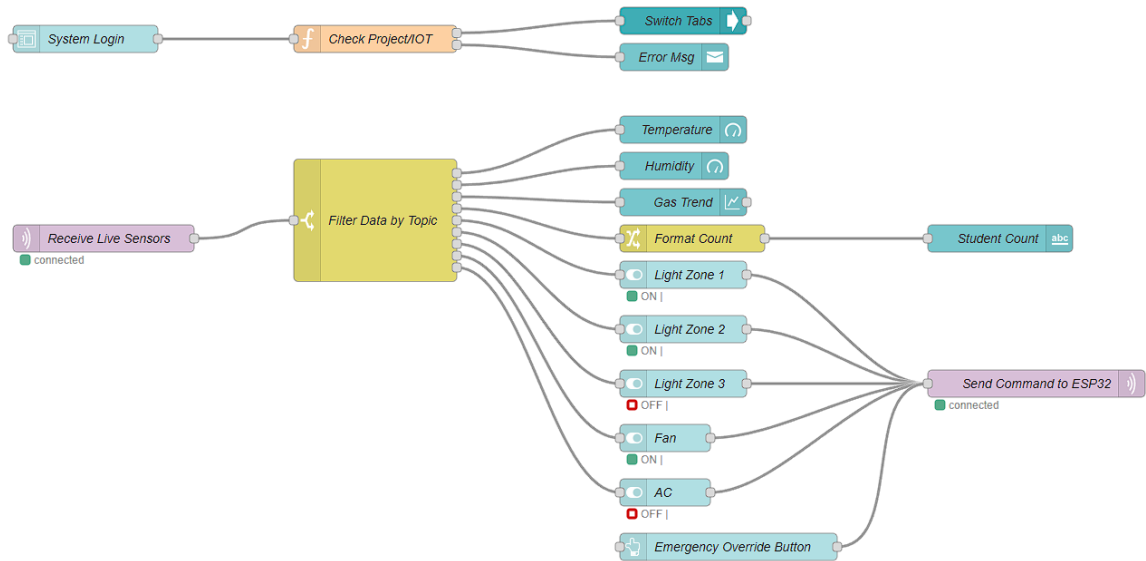
Live Gas Level: 3295

Live Gas Level: 3497

Live Gas Level: 3700

Live Gas Level: 3915

Node-RED:



≡ Login Tab

Authentication

Enter Credentials

Username *

Project

Password *

LOGIN

